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PROTOCOL

Antineoplastic Effects of Petiveria alliacea: A Systematic Review Protocol

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Abstract

Petiveria alliacea L. (*Phytolaccaceae*), commonly known as "Anamú" or "Guinea Hen Weed," is a medicinal plant traditionally used in herbal medicine across Latin America, the Caribbean, and Africa. Preclinical investigations have attributed cytotoxic, antioxidant, immunomodulatory, and proapoptotic properties to its bioactive compounds, indicating potential antineoplastic activity. However, available evidence remains fragmented across different experimental models, cancer types, plant preparations, and study designs, and no systematic synthesis of the antineoplastic evidence base has been published to date. This protocol describes the methodology for a systematic review aiming at evaluating the antineoplastic effects of *Petiveria alliacea* in *in vitro* cell models, *in vivo* animal studies, and clinical trials involving cancer patients. Searches will be conducted in ten electronic databases (including MEDLINE, EMBASE, Scopus, LILACS, Web of Science, ScienceDirect, SciELO, Epistemonikos, ClinicalTrials.gov, and EudraCT) in English, Spanish, or Portuguese, covering studies published from January 2000 onwards. Eligible studies will include controlled *in vitro*, *in vivo*, and clinical investigations reporting antineoplastic outcomes. Study selection will be performed independently by two reviewers using Rayyan. Risk of bias will be assessed using validated tools for preclinical research. Subgroup and sensitivity analyses are planned by tumor type, extract type, plant part, and study model. This protocol was registered in PROSPERO on 26 March 2026 (registration number: CRD420251108791).

Keywords: antineoplastic agents; cancer; complementary therapies; *petiveria alliacea*.

1. Introduction

Cancer represents one of the leading causes of morbidity and mortality worldwide, with an estimated 20 million new cases and nearly 10 million deaths recorded globally in 2022 [1]. The burden is disproportionately concentrated in low- and middle-income countries, where access to surgery, radiotherapy, and innovative systemic therapies remains severely limited [2]. In this context, plant-derived medicinal products play a prominent role as primary or complementary health resources across diverse cultures, and represent an important reservoir of bioactive compounds for pharmaceutical research [3].

Petiveria alliacea L. (*Phytolaccaceae*) (known as "Anamú," "Guinea Hen Weed," "mucura," or "tipi") is a perennial herb indigenous to tropical and subtropical regions of Latin America and the Caribbean, with distribution also documented in Africa. Ethnobotanical records document its widespread traditional use for anti-inflammatory, analgesic, antimicrobial, and immunostimulatory purposes [4, 5]. Its bioactive constituents include organosulfur compounds,

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coumarins, benzaldehyde derivatives, flavonoids, and thiazolidine compounds [6]. Preclinical investigations have associated *Petiveria alliacea* extracts with inhibition of cancer cell proliferation, induction of apoptosis, antioxidant activity, and reduction of tumor growth in animal models [7,8]. Despite accumulating evidence, investigations have been conducted across heterogeneous cancer types, experimental systems, plant preparations, and outcome measures, preventing the drawing of unified conclusions across studies.

To date, no systematic review has comprehensively synthesised the antineoplastic evidence for *Petiveria alliacea*. This absence limits the ability to identify consistent activity patterns, assess methodological rigor, and guide translational research priorities. This protocol describes the methods for a systematic review aiming at closing this gap, contributing to the evidence base for plant-derived antineoplastic agents and informing future research in integrative oncology.

2. Methods

2.1. Aim and objectives

This systematic review aims to evaluate the antineoplastic effects of *Petiveria alliacea* in *in vitro* models, *in vivo* animal studies, and clinical trials. The specific objectives are:

- To assess the cytotoxic and antioxidant/prooxidant activity of *Petiveria alliacea* preparations in *in vitro* neoplastic cell models, as measured by the inhibitory cytotoxic concentration (IC_{50}) and validated antioxidant assays.
- To evaluate the antineoplastic efficacy of *Petiveria alliacea* in controlled *in vivo* murine tumor models, based on survival outcomes, treatment response, and tumor suppression.
- To summarize available evidence from clinical trials evaluating *Petiveria alliacea* in cancer patients, including disease-free survival, progression-free survival, overall survival, treatment response rate, and the secondary clinical endpoints: safety and toxicity profiles and health-related quality of life.
- To assess the risk of bias and methodological quality of all included studies.
- To characterize heterogeneity in *Petiveria alliacea* preparations and cancer models across included studies, and to identify gaps in the evidence base.

2.2. Study design

This review will follow the methodological guidance for systematic reviews of pre-clinical intervention studies, incorporating the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 framework. Risk of bias assessment is based on validated tools for preclinical research. The protocol was registered prospectively in PROSPERO on 26 March 2026 (CRD420251108791) [9].

2.3. Eligibility criteria

2.3.1. Population and models

In vitro studies using established neoplastic cell lines of any type are eligible. *In vivo* studies in murine models (mice or rats) bearing experimentally induced solid or hematological tumors, with housing conditions and ethical approval, will be included. Clinical studies enrolling adult

patients (≥ 18 years) with confirmed cancer diagnosis of any type or stage will also be eligible. Studies using non-murine animal species, healthy animals without neoplastic disease, or pediatric populations will be excluded.

2.3.2. Intervention

The intervention of interest is the administration of *Petiveria alliacea* in any preparation (whole plant, aqueous extract, ethanolic extract, methanolic extract, infusion, or volatile extract), and at any dose. Variable administration routes, frequency, and treatment durations are also acceptable. Studies combining *Petiveria alliacea* with other agents will be included only when the effects of *Petiveria alliacea* could be evaluated independently.

2.3.3. Comparator

Eligible comparators included vehicle control, untreated cells or animals, placebo, no intervention, or standard antineoplastic therapy. Single-arm studies without a control group will be excluded.

2.3.4. Outcomes

Primary outcomes for *in vitro* studies are IC_{50} (μg/mL or mg/mL) and antioxidant or prooxidant activity measured by validated assays (ABTS, DPPH, FRAP). Primary outcomes for *in vivo* and clinical studies are disease-free survival, progression-free survival, overall survival, treatment response rate, and presence of distant metastases. Secondary outcomes included induction of immunogenic cell death markers (ATP depletion, calreticulin expression, HMGB1 release) and changes in metabolic cellular profile. For clinical studies, secondary outcomes also include safety and toxicity profiles (incidence, type, and severity of adverse events) and health-related quality of life, assessed with validated instruments. Studies that do not report any primary antineoplastic outcome will be excluded.

2.3.5. Study design

Randomized controlled trials, non-randomized controlled studies, controlled preclinical studies (*in vitro* and *in vivo*), semi-experimental animal studies, and controlled clinical cohort studies are eligible. Only original peer-reviewed research articles published from January 2000 onwards in English, Spanish, or Portuguese will be included. Case reports, case series, narrative reviews, scoping reviews, conference abstracts, editorials, letters to the editor, theses, and *in silico* studies without experimental validation are excluded.

2.4. Search strategy

Searches will be conducted in the following electronic databases: MEDLINE (via PubMed), EMBASE (via Embase.com), Scopus, Web of Science Core Collection, LILACS (via BVS), ScienceDirect, SciELO, Epistemonikos, ClinicalTrials.gov, and EudraCT. Grey literature will be sought via Google Scholar, Trip Medical Database, Latin American academic repositories, and manual reference checking of included studies. Consistent with the eligibility criteria, only studies published in English, Spanish, or Portuguese will be included; this pragmatic restriction reflects the working languages of the review team and is acknowledged as a minor limitation of the review. Studies published from January 2000 to the date of search will be eligible. This close date limit was adopted because the modern, peer-reviewed body of preclinical research on

Petiveria alliacea's antineoplastic activity began to emerge around the year 2000, and because this cut-off improves alignment with contemporary methodological and reporting standards, preserving full-text record retrievability.

Searches were conducted following the PRESS checklist and structured around (i) *Petiveria alliacea* and its known synonyms combined with controlled vocabulary (MeSH, Emtree, DeCS); and (ii) antineoplastic and cancer-related terms. Strategies were adapted to each database's syntax and controlled vocabulary. Complete executable search strings, database names, access platforms, search fields, and settings are provided in Supplementary Material (Suppl. 1).

2.5. Study selection

The selection of studies will be conducted in three sequential phases: title screening, abstract screening, and full-text review. Each phase will be performed independently by two reviewers using the Rayyan® software [10], with bibliographic fields blinded during title and abstract screening. Discrepancies will be resolved by consensus, and a third reviewer will adjudicate unresolved disagreements. Screening decisions and exclusion reasons will be documented according to the PRISMA 2020 flow reporting guideline. Before screening, all records retrieved from the electronic databases will be imported into a reference manager and into Rayyan®, where duplicate records will be removed automatically and then verified manually. When two or more publications report the same experiment or the same patient cohort (companion or redundant reports), the most complete and most recent report will be used as the primary data source and the remaining reports will be linked to it, so that no outcome is counted more than once.

2.6. Data extraction

Data extraction will be performed independently by four reviewers using a standardized Microsoft Excel template. At least two reviewers in parallel will extract each study; discrepancies will be resolved by consensus or third-party adjudication. Authors will be contacted by email when essential data are missing or unclear.

Variables to be extracted include: bibliographic details; study design and structure; population or model characteristics; intervention details (preparation form, plant part, origin, extraction method, dose, route, frequency, duration); comparator description; and all primary and secondary outcomes with associated statistical measures (mean $\pm$ SD, IC_{50}, hazard ratio, confidence interval, p-value). Final data files will be deposited in an open repository upon review completion.

2.7. Risk of bias assessment

Bias risk will be assessed independently by three reviewers. For *in vivo* animal studies, the SYRCLE risk of bias tool [11] and the CAMARADES checklist [12] will be applied. For *in vitro* studies, the methodological rigor and reporting completeness of each included study will be appraised using the Materials Design, Analysis and Reporting (MDAR) framework and checklist [14], covering key elements such as cell-line authentication, biological and technical replication, statistical reporting, and reagent traceability. For clinical studies, the Cochrane Risk of Bias tool version 2 will be applied to randomized trials, and the Newcastle-Ottawa Scale to non-randomized studies. A fourth reviewer will arbitrate persistent disagreements. Results will be summarized narratively and presented graphically.

2.8. Data synthesis

A narrative synthesis will be conducted for all outcomes, providing descriptive summaries and visual representations of the evidence. The pooling of studies will be based on clinical and methodological comparability and assessed via the I^2 statistic. Random-effects models will be used for pooling. Subgroup analyses will be planned by tumor type, extract type, plant part used, and study model. Sensitivity analyses will integrate the exclusion of studies at high risk of bias and of non-randomized studies, where applicable. Potential publication bias will be addressed qualitatively; formal statistical tests will be planned only if 10 or more studies are available for an outcome in a meta-analysis.

To account for the substantial chemical heterogeneity of herbal products, *Petiveria alliacea* preparations will be grouped *a priori* according to predefined criteria, extraction solvent and polarity (aqueous/polar versus organic/non-polar), plant part used, and whole-plant versus fractionated or standardized extract, and outcomes will be summarized separately within these predefined categories rather than pooled across them. Quantitative pooling will not be undertaken when studies differ in these core biological dimensions (different extract or solvent classes, plant parts, tumor lineages, or outcome definitions), even when statistical heterogeneity (I^2) appears acceptable, because pooling across biologically non-comparable conditions may generate misleading estimates. When a meta-analysis is not appropriate or feasible, the evidence will be synthesized narratively following the Synthesis Without Meta-analysis (SWiM) reporting guideline [15].

3. Discussion

This systematic review protocol describes a rigorous, pre-registered methodology to synthesize the antineoplastic evidence for *Petiveria alliacea* across experimental and clinical settings. To the best of the review team's knowledge, this is the first systematic review with a prospectively registered protocol addressing this question, filling a recognized gap in the integrative oncology and phytomedicine literature.

The global burden of cancer disproportionately affects low- and middle-income countries where plant-based medicines are widely used, justifying the systematic evaluation of ethnobotanically relevant plants with preclinical antineoplastic signals [1–3]. *Petiveria alliacea* is a compelling candidate: documented traditional use, partially characterized phytochemistry, and substantive preclinical data suggesting cytotoxic and immunomodulatory mechanisms [7, 8]. The absence of a synthesized, critically appraised evidence base prevents rational prioritization of translational research or the development of evidence-based guidance for clinicians and regulators.

The methodological strengths of this review include: a prospective PROSPERO registration, a comprehensive multi-database search across ten electronic databases, an independent dual screening with bibliographic blinding, a validated preclinical risk of bias, the use of reporting tools (SYRCLE, CAMARADES, and MDAR), and pre-specified subgroup analyses aligned with known sources of heterogeneity.

The anticipated limitations of this work include publication bias toward positive results in preclinical oncology literature [13], heterogeneity in *Petiveria alliacea* preparations and cancer models, the restriction of eligible studies to three languages (English, Spanish, and Portuguese), which introduces a degree of language bias, and the probable paucity of clinical evidence. These limitations are addressed through grey literature searching, qualitative synthesis of design

variability, and transparent reporting. The declared conflict of interest of one member of the review team is mitigated through independent dual review by authors without commercial interests, as detailed in the Conflict-of-Interest statement.

The findings of this review are expected to provide the first consolidated, critically appraised evidence map for the antineoplastic properties of *Petiveria alliacea*, identify priority cancer types and preparation strategies for future investigation, and contribute to evidence-based frameworks for plant-derived antineoplastic agents in Latin America and globally.

4. Conclusion

As this is a protocol, no definitive conclusions can be drawn at this stage. The planned systematic review will provide a rigorous, comprehensive synthesis of the antineoplastic evidence for *Petiveria alliacea* across in vitro, in vivo, and clinical study designs, assessing the methodological quality of the existing literature, and generating specific recommendations for future research in integrative oncology.

5. Acknowledgments

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6. Conflict of interest

Ricardo Ballesteros-Ramírez discloses a partnership in DreemBio, a company that holds patents related to *Petiveria alliacea*. This potential conflict of interest was declared prospectively in the PROSPERO registration (CRD420251108791). All study selection, data extraction, and risk of bias assessments will be conducted independently by review team members without commercial interests; final decisions will require consensus of two or more independent reviewers. The remaining authors declare no conflicts of interest.

7. AI Use Disclosure Statement

The authors used ChatGPT, developed by OpenAI, during the preparation of this manuscript. The tool was used exclusively for language assistance, specifically for English language editing, grammar and syntax correction, style and clarity improvement, and translation support from Spanish into English. All outputs generated by the AI tool were carefully reviewed, verified, and edited by the authors. The authors take full responsibility for the accuracy, integrity, and originality of all content presented in this manuscript.

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Efectos antineoplásicos de *Petiveria alliacea*: Protocolo de revisión sistemática

Resumen: *Petiveria alliacea* L. (*Phytolaccaceae*), conocida comúnmente como "Anamú" o "Guinea Hen Weed", es una planta medicinal utilizada tradicionalmente en medicina herbolaria en América Latina, el Caribe y África. Investigaciones preclínicas han atribuido propiedades citotóxicas, antioxidantes, inmunomoduladoras y proapoptóticas a sus compuestos bioactivos, lo que indica una potencial actividad antineoplásica. Sin embargo, la evidencia disponible permanece fragmentada entre diferentes modelos experimentales, tipos de cáncer, preparaciones vegetales y diseños de estudio, y hasta la fecha no se ha publicado ninguna síntesis sistemática de la base de evidencia antineoplásica. Este protocolo describe la metodología para una revisión sistemática orientada a evaluar los efectos antineoplásicos de *Petiveria alliacea* en modelos celulares *in vitro*, estudios en animales *in vivo* y ensayos clínicos con pacientes con cáncer. Las búsquedas se realizarán en diez bases de datos electrónicas (incluyendo MEDLINE, EMBASE, Scopus, LILACS, Web of Science, ScienceDirect, SciELO, Epistemonikos, ClinicalTrials.gov y EudraCT) en inglés, español o portugués, abarcando estudios publicados a partir de enero de 2000. Los estudios elegibles incluirán investigaciones controladas *in vitro*, *in vivo* y clínicas que reporten resultados antineoplásicos. La selección de estudios será realizada de forma independiente por dos revisores mediante Rayyan. El riesgo de sesgo se evaluará con herramientas validadas para investigación preclínica. Se planea hacer análisis de subgrupos y de sensibilidad por tipo de tumor, tipo de extracto, parte de la planta y modelo de estudio. Este protocolo fue registrado en PROSPERO el 26 de marzo de 2026 (número de registro: CRD420251108791).

Palabras Clave: Agentes antineoplásicos; Cáncer; *Petiveria alliacea*; Terapias complementarias

Efeitos antineoplásicos de *Petiveria alliacea*: Protocolo de revisão sistemática

Resumo: *Petiveria alliacea* L. (*Phytolaccaceae*), comumente conhecida como "Anamú" ou "Guinea Hen Weed", é uma planta medicinal tradicionalmente utilizada na medicina herbolária na América Latina, no Caribe e na África. Investigações pré-clínicas atribuíram propriedades citotóxicas, antioxidantes, imunomoduladoras e pró-apoptóticas aos seus compostos bioativos, indicando potencial atividade antineoplásica. No entanto, as evidências disponíveis permanecem fragmentadas entre diferentes modelos experimentais, tipos de câncer, preparações vegetais e delineamentos de estudo, e até o momento nenhuma síntese sistemática da base de evidências antineoplásicas foi publicada. Este protocolo descreve a metodologia para uma revisão sistemática com o objetivo de avaliar os efeitos antineoplásicos de *Petiveria alliacea* em modelos celulares *in vitro*, estudos em animais *in vivo* e ensaios clínicos envolvendo pacientes com câncer. As buscas serão realizadas em dez bases de dados eletrônicas (incluindo MEDLINE, EMBASE, Scopus, LILACS, Web of Science, ScienceDirect, SciELO, Epistemonikos, ClinicalTrials.gov e EudraCT) em inglês, espanhol ou português, cobrindo estudos publicados a partir de janeiro de 2000. Os estudos elegíveis incluirão investigações controladas *in vitro*, *in vivo* e clínicas que relatem resultados antineoplásicos. A seleção dos estudos será realizada de forma independente por dois revisores utilizando o Rayyan. O risco de viés será avaliado por meio de ferramentas validadas para pesquisa pré-clínica. Planeja-se fazer análises de subgrupo e de sensibilidade por tipo de tumor, tipo de extrato, parte da planta e modelo de estudo. Este protocolo foi registrado no PROSPERO em 26 de março de 2026 (número de registro: CRD420251108791).

Palavras-chave: Agentes antineoplásicos; Câncer; Petiveria alliacea; Terapias complementares

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